

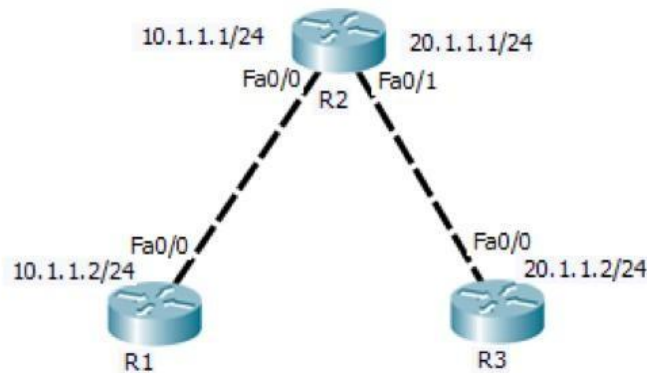
DAY-4

Introduction to OSPF Theory : How OSPF works: Areas, LSAs, and cost . Practical: Observe OSPF adjacency formation in Packet Tracer. Task: Write down steps to configure OSPF in a network.

Q-1 Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.

[A]

- Network Topology



Q-2 Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.

[A]

- Assign IP :

Router	Interface	IP Address	Subnet Mask
R1	FastEthernet0/0	10.1.1.2	255.255.255.0
R2	FastEthernet0/0	10.1.1.1	255.255.255.0
R2	FastEthernet0/1	20.1.1.1	255.255.255.0
R3	FastEthernet0/0	20.1.1.2	255.255.255.0

- PING Command : From Router R1 ping 10.1.1.1

Result: Success (100% success rate)

Q-3 Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks.

[A]

- > **Router R1 Configuration** enable
configure terminal

```
router ospf 1 network 10.1.1.0  
0.0.0.255 area 0
```

- **Router R2 Configuration** enable
configure terminal

```
router ospf 1 network 10.1.1.0  
0.0.0.255 area 0 network 20.1.1.0  
0.0.0.255 area 0
```

- **Router R3 Configuration** enable
configure terminal

```
router ospf 1 network 20.1.1.0  
0.0.0.255 area 0
```

- **Verify OSPF Configuration**

Run the following command on all three routers:

show ip protocols Expected

Output:

- Routing Protocol: **OSPF**
- Process ID: **1**
- Area: **0**
- Advertised Networks:
 - **R1:** 10.1.1.0/24 ○ **R2:** 10.1.1.0/24, 20.1.1.0/24
 - **R3:** 20.1.1.0/24

Q-4 Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router. Hint : Look for the FULL/DR or FULL/BDR state to confirm adjacency.

[A]

➤ **Command**

Run the following command on each router:
show ip ospf neighbor

➤ **Router R1**

R1# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:39	10.1.1.1	FastEthernet0/0

➤ **Router R2**

R2# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/BDR	00:00:36	10.1.1.2	FastEthernet0/0
3.3.3.3	1	FULL/DR	00:00:34	20.1.1.2	FastEthernet0/1

➤ **Router R3**

R3# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/BDR	00:00:38	20.1.1.1	FastEthernet0/0

➤ **Observation**

Router	Neighbor	State	Interface
R1	R2	FULL/DR	FastEthernet0/0
R2	R1	FULL/BDR	FastEthernet0/0
R2	R3	FULL/DR	FastEthernet0/1
R3	R2	FULL/BDR	FastEthernet0/0

Q-5 Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table. Hint : Use show ip route ospf to see route changes after modifying the cost.

[A]

> Step 1: Change the OSPF Cost

On **Router R2**, increase the OSPF cost of the interface connected to **R3**.

R2> enable

R2# configure terminal

R2(config)# interface FastEthernet0/1

R2(config-if)# ip ospf cost 100

R2(config-if)# end

➤ Step 2: Verify the OSPF Routing Table

Run the following command on **Router R1**:

show ip route ospf

➤ Sample Output

R1# show ip route ospf

O 20.1.1.0/24 [110/101] via 10.1.1.1, FastEthernet0/0 Run
the same command on **Router R3**:

R3# show ip route ospf

O 10.1.1.0/24 [110/101] via 20.1.1.1, FastEthernet0/0

➤ Observation Table

Before Changing Cost	After Changing Cost
OSPF cost was the default value (1).	OSPF cost increased to 100 on FastEthernet0/1.
Route metric was lower.	Route metric became higher (101).
Traffic used the only available path through R2.	Traffic still used the same path because no alternate route exists.
OSPF routing table showed a lower metric.	OSPF routing table showed a higher metric.